ATREYU SUTTON

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SUMMARY

M.S. student at CU Boulder (B.A. Computer Science, 4.0 GPA) who builds LLM-powered systems and machine learning models end to end: autonomous agents, voice assistants, schema-validated generation pipelines and demand forecasting in Python and TypeScript.

EDUCATION

M.S. Engineering, Creative Technology and Design *August 2025 to May 2027*
University of Colorado Boulder, Boulder, CO

B.A. Computer Science, 4.0 GPA *August 2020 to May 2023*
Rollins College, Winter Park, FL

Coursework: RISD (Industrial Design Sketching & Rendering), UMass Amherst (Sustainable Engineering & Architecture) *2019*

AI PROJECTS

Fantasia: Autonomous Nightly Build Agent (fantasia.build) *September 2026 to Present*
Claude Code, Python, Bash, launchd, GitHub CLI, Tailscale

- Built an unattended agent on a dedicated Mac Mini that each night researches new AI releases, weighs about eight candidate ideas, then builds, tests and publishes one project with a live demo.
- Shipped 50 public repositories since September 2026 (one per night on 45 of the last 47 nights), each publishing the agent's reasoning.
- Designed the safety model for running an agent without approval prompts: an isolated non-admin account, a separate GitHub identity, a command guard that blocks destructive shell and git operations, and an SSH-only network policy.

Lesson Desk: AI Lesson Builder for a Physics Professor *November 2026*
TypeScript, Next.js, Claude API, Zod, Python (PyMuPDF), Cloudflare Workers, D1

- Built a web app that turns a one-line request into a 90-minute lesson grounded in the course textbook: slides, a group activity, a worksheet with a page-cited answer key, and two shuffled test versions.
- Wrote a Python pipeline that splits textbook PDFs by chapter and extracts and indexes 1,119 figures, with pixel-level verification.
- Designed a multi-stage generation pipeline (route, plan, then parallel generation) with schema-validated structured outputs and sanitization that rejects invented figure references; student names never reach the model.

TorQ: Voice Agent for Car Repair on a Tracked Robot *September 2026 to Present*
Python, Claude API, faster-whisper, Piper TTS, Raspberry Pi 5, Arduino

- Built the voice assistant for a tracked garage robot: local Whisper transcription, Claude answers, and interruptible spoken replies.
- Implemented a streaming tool-use loop (vehicle record, job notes, motor stop) with routing between a fast and a stronger model, prompt caching and persistent job memory.
- Wrote drive firmware with a watchdog and emergency stop, plus 18 integration tests against simulated hardware and model responses.

FuelFed Motor Market: Collector Car Marketplace *July 2025 to April 2026*
TypeScript, React, Next.js, PostgreSQL (Supabase, Prisma), Cloudflare Images

- Sole engineer on a marketplace for European collector cars; later rewrote it in Next.js and TypeScript on a 9-model PostgreSQL schema.
- Built bulk-import pipelines that parse, validate, normalize and deduplicate vehicle data, and wrote the SQL migration for full-text search.

EXPERIENCE

Founder & Lead Developer *January 2023 to Present*
Sutton Web Solutions, Boulder, CO

- A bakery (NDA): built a demand-forecasting model (LightGBM, Poisson and quantile objectives) on 1.1M sales records that predicts daily pastry demand per store and item three days ahead from lag, holiday and weather features; in walk-forward backtests it cut forecast error (WAPE) from 33.8% for a last-week baseline to 25.1%.
- Web platforms for clients: FuelFed Motor Market (online collector car marketplace), Valdra Outside (outdoor brand site), Ute Pass Vacation Rentals and The Real Estate Collaborative (real estate), and Nest Messages (communications).

SKILLS

AI & LLM Engineering: LLM application development (Claude API), agentic workflows and tool use, context engineering, structured outputs and schema validation, prompt caching, model routing, speech pipelines (Whisper, Piper)

Machine Learning: Demand forecasting with gradient-boosted trees (LightGBM), feature engineering, walk-forward backtesting, pandas

Agent Operations: Claude Code automation, autonomous build-and-verify pipelines, sandboxing and guardrails, cost and time budgeting

Programming & Data: Python, TypeScript, SQL (PostgreSQL), PDF and data preprocessing pipelines, Git

Mechanical & Hardware: Robotics prototyping (Raspberry Pi, Arduino), CAD design, 3D modeling, engine diagnostics

Professional: Communication, leadership, problem solving, team collaboration